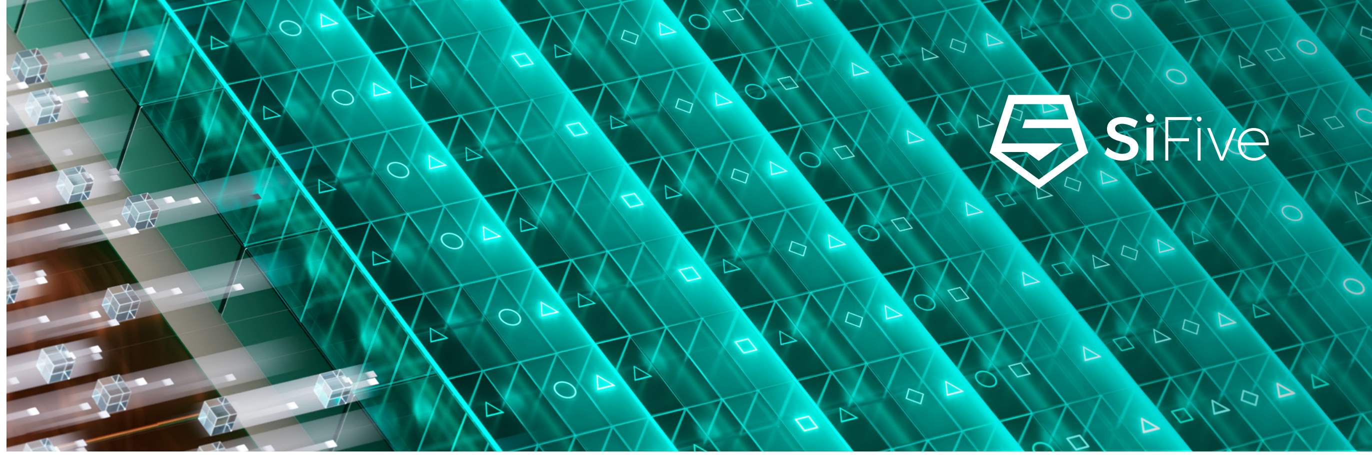
PRODUCT BRIEF

INTELLIGENCE FAMILY Gen 2



SiFive

PRODUCT OVERVIEW

The SiFive® Second Generation Intelligence™ family of products are designed for efficient execution and maximum throughput of AI workloads from the far edge to the cloud. Combining scalar, vector and matrix computing with a standard RISC-V software stack has never been easier. Whether you are looking for something to bring vector compute to the edge, or work together with your custom accelerator, or you need a full solution with a scalable matrix engine – we have you covered.

Built around a dual issue 8-stage in-order superscalar pipeline, each product supports multi-core/multi-cluster configurations and a wide variety of datatypes, making it possible to support diverse requirements across many applications. With dedicated, loosely coupled vector pipelines to efficiently process AI/ML workloads without negative impacts of memory latency-based stalls typical of most other CPUs. Co-processor use cases are also heavily optimized thanks to the specialized SiFive Scalar Coprocessor Interface (SSCI) and Vector Coprocessor Interface eXtension (VCIX) interfaces which enabling direct connectivity between the processor core and a custom hardware accelerator.

APPLICATIONS & USES

The highly configurable Intelligence product family can deliver from GOPs to thousands of TOPs and therefore can serve in a broad market from deeply embedded edge AI processors to automotive and aerospace, to the data center.

KEY FEATURES

- Scalar, Vector & Matrix Processing
- SiFive Intelligence Extensions
- SSCI & VCIX Accelerator Control Interfaces
- Highly Scalable and Configurable
- RVA23 Support
- Power Efficiency for AI/ML

MAJOR BENEFITS

- High data throughput and memory bandwidth up to 1TB/s
- Highly configurable from lowest area to highest TOPs
- Architected to tolerate and reduce system level memory latencies
- High bandwidth and low latencies interfaces to custom accelerators

Check out SiFive's Intelligence 2nd Gen products @SiFive.com/cores/intelligence or contact us directly for more information

Contact Sales

PRODUCT BRIEF

INTELLIGENCE FAMILY Gen 2

		X-Series, 100 Class		X-Series, 200 Class		X-Series, 300 Class	XM Series
		X160 Gen 2	**X180 Gen 2**	**X288 Gen 1**	**X280 Gen 2**	**X390 Gen 2**	**XM Gen 2**
	Base ISA	RV32I(E)	RV64I	RV64	RVA23	RVA23	Highly scalable and configurable Matrix engine clusters that include 4 X300 Gen 2 class Scalar and Vector CPUs. Contact Sales for more information. Contact Sales for more information
	# Cores per Cluster	Up to 4		1, 2, 4		1, 2, 4	
	Decode width	2		2		2	
	#pipeline stages	8		8		8	
	FPU Support	SP, DP, HP		SP, DP, HP		SP, DP, HP	
	Vector VLEN	128-bit		512-bit		1024-bit	
	Vector Datatype	INT8, INT16, INT32, INT64, BF16, FP16, FP32		INT8, INT16, INT32, INT64, BF16, FP16, FP32, FP64	INT8, INT16, INT32, INT64, FP4, FP8, BF16, FP16, FP32, FP64 (OFP4 and OFP8 via conversion)		
	Hardened for Aerospace (DCLS etc)	No		Yes	No	No	
	Debug/Trace	Yes		Yes		Yes	
	AIA Interrupts	Yes		Yes		Yes	
	HPSC features incl. DCLS	No		Yes	No	No	
Memory	**L1 ITIM**	0 – 2 MiB		4KiB		4KiB	
Memory	**L1 I-Cache**	1 – 64KiB,		32KiB	8KiB- 64KiB	8KiB- 64KiB	
Memory	**L1 dTIM**	0 – 2 MiB		No		No	
Memory	**L1 d-Cache**	1 – 256KiB		64KiB	8KiB- 64KiB	8KiB- 64KiB	
Memory	**L2 Private Cache**	No		256KiB	No	No	
Memory	**L2 Shared Cache**	0 – 4MiB		No	Up to 4MiB	Up to 4MiB	
Memory	**L3 Shared Cache**	No		Up to 8MB	No	No	
Memory	**MMU**	No		sv48	sv48, sv57	sv48, sv57	
Ports	**VCIX**	64, 128-bit		No	1024-bit	2048-bit	
Ports	**SSCI**	Yes		No	Yes	Yes	
Ports	**Core Local Port** (Scales per Core)	32 – 128-bit		No	256-bit	512-bit	
Ports	**Core Local Front Port**	No		No		No	
Ports	**System Port**	32 – 128-bit		64-bit, 128-bit	No	No	
Ports	**Memory Port**	32 – 128-bit		128-bit, 256-bit	Up to 2x 256-bit	Up to 2x 512-bit	
Ports	**Peripheral Port**	32 – 128-bit	32 – 64-bit	64-bit		64-bit	
Ports	**Control Port**	No		No	64-bit	64-bit	
Ports	**Front Port**	32 – 128-bit		2x 256-bit	256-bit	256-bit	